Math 1040 **EXAM #1**

Name KEY
(print)

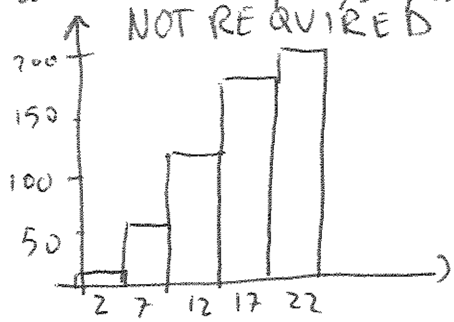
Total = 100 points Please show all your work.

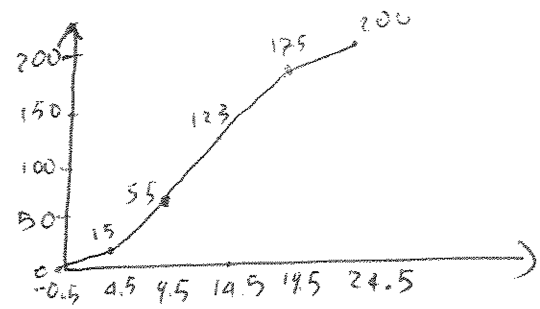
1. (20 points)

Class	Frequency, f	CUMULATIVE FREQUENCY	MIDPOINT
0-4	15	15	2
5-9	40	55	7
10-14	68	123	12
15-19	52	175	17
20-24	25	200	22

a) Use the given frequency distribution to construct a cumulative frequency distribution (you can continue writing in the table above) and an ogive (cumulative frequency graph).

cumulative freq. hystogram
NOT REQUIRED





b) Approximate the mean of the grouped data.

$$\bar{X} = \frac{15\times2+40\times7+68\times12+52\times17+25\times22}{200} = 12.8$$

Mean 12.8

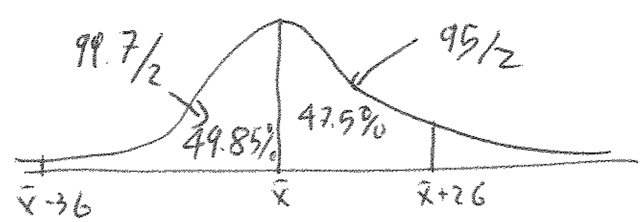
2. (15 pts) You are applying for a job and the company that you are considering has a mean employee salary of \$ 37,000 with a standard deviation of \$ 1,500. If the company employs 650 people, use the Empirical Rule to determine about how many employees earn between \$ 32,500 and \$ 40,000?

$32,500 = 37,000 - 4500 = \bar{x} - 3\sigma$

$40,000 = 37000 + 3,000 = \bar{x} + 2\sigma$

$49.85 + 47.5 = 97.35$

$.9735 \times 650 \approx 633$

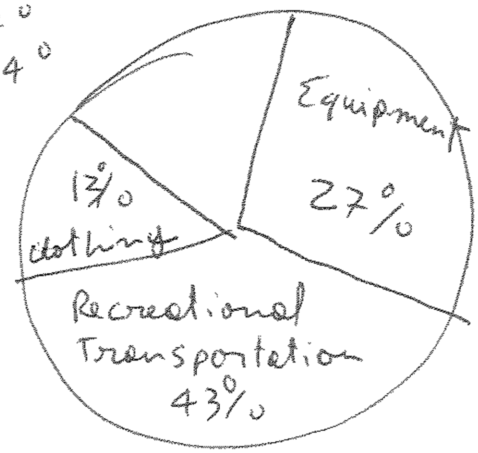


Answer 633

3. (16 pts) Display the data below using a Pie Chart (clearly indicate the percentages that correspond to each category):

U.S. sporting goods sales (in billions of dollars) can be classified in four areas:

Areas	\$ (in billions)	rel freq	angle
Clothing	11.7	13%	47°
Footwear	15.7	17%	61°
Equipment	24.0	27%	97°
Recreational-transport	38.5	43%	154°
	89.9		



4. (20 pts) In a random sample, 60 people were asked how many times per year they go to the movie theater. Approximate the sample standard deviation (hint: first you need to find the midpoints)
These are the results:

class (times per year)	frequency	Midpoints	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
0-4	6	2	-7.33	53.73
5-9	28	7	-2.33	5.43
10-14	18	12	2.67	7.13
15-19	8	17	7.67	58.83

$$\bar{x} = \frac{2\times 6 + 7\times 28 + 12\times 18 + 17\times 8}{60} = 9.33$$

$$SS_x = 1073.40 \qquad s = \sqrt{\frac{SS_x}{n-1}} = \sqrt{\frac{1073.40}{59}} = 4.27$$

Mean 9.3

Standard deviation 4.27

5. (15 pts) The mean sale per customer for 70 customers at a gas station is $42, with a standard deviation of $4.50. On the basis of Chebychev's Theorem, at least how many of the customers spent between $33 and $51?

$33 = 42 - 9 = \bar{x} - 2\sigma$

$51 = 42 + 9 = \bar{x} + 2\sigma$

$k = 2 \qquad 1 - \frac{1}{k^2} = 1 - \frac{1}{4} = .75$

$.75 \times 70 = 52.5$

Answer 52 (or 53)

6. (6 pts) Identify which sampling technique was used in the study (random, cluster, stratified, systematic, convenience):

a) Security agents at the airport stopped every 15th person for a detailed search. systematic

b) 20 students are randomly selected from each grade level at an elementary school and surveyed about their study habits. stratified

c) A student asks 20 friends to participate in a sociology class experiment. convenience

7. (8 pts) Identify the data set's level of measurement (nominal, ordinal, interval, ratio):

a) the average daily temperature (in degrees Fahrenheit) on ten randomly selected days interval

b) the size-class for a sample of automobiles listed as:
subcompact compact midsize full size ordinal

c) ages of students in a statistics class ratio

d) the years of birth for runners in a marathon interval